

4.14 Classwork: Normal distribution

3. [Maximum mark: 6]

A company produces bags of sugar whose masses, in grams, can be modelled by a normal distribution with mean 1000 and standard deviation 3.5. A bag of sugar is rejected for sale if its mass is less than 995 grams.

- (a) Find the probability that a bag selected at random is rejected. [2]
- (b) Estimate the number of bags which will be rejected from a random sample of 100 bags. [1]
- (c) Given that a bag is not rejected, find the probability that it has a mass greater than 1005 grams. [3]



4. [Maximum mark: 14]

It is known that the weights of male Persian cats are normally distributed with mean 6.1 kg and variance 0.5^2 kg^2 .

(a) Sketch a diagram showing the above information. [2]

(b) Find the proportion of male Persian cats weighing between 5.5 kg and 6.5 kg. [2]

A group of 80 male Persian cats are drawn from this population.

(c) Determine the expected number of cats in this group that have a weight of less than 5.3 kg. [3]

(d) It is found that 12 of the cats weigh more than x kg. Estimate the value of x . [3]

(e) Ten of the cats are chosen at random. Find the probability that exactly one of them weighs over 6.25 kg. [4]

Do **not** write solutions on this page.

8. [Maximum mark: 15]

The flight times, T minutes, between two cities can be modelled by a normal distribution with a mean of 75 minutes and a standard deviation of σ minutes.

- (a) Given that 2% of the flight times are longer than 82 minutes, find the value of σ . [3]
- (b) Find the probability that a randomly selected flight will have a flight time of more than 80 minutes. [2]
- (c) Given that a flight between the two cities takes longer than 80 minutes, find the probability that it takes less than 82 minutes. [4]

On a particular day, there are 64 flights scheduled between these two cities.

- (d) Find the expected number of flights that will have a flight time of more than 80 minutes. [3]
- (e) Find the probability that more than 6 of the flights on this particular day will have a flight time of more than 80 minutes. [3]



Do **not** write solutions on this page.

9. [Maximum mark: 16]

The random variable X follows a normal distribution with mean μ and standard deviation σ .

(a) Find $P(\mu - 1.5\sigma < X < \mu + 1.5\sigma)$. [3]

The avocados grown on a farm have weights, in grams, that are normally distributed with mean μ and standard deviation σ . Avocados are categorized as small, medium, large or premium, according to their weight. The following table shows the probability an avocado grown on the farm is classified as small, medium, large or premium.

Category	Small	Medium	Large	Premium
Probability	0.04	0.576	0.288	0.096

The maximum weight of a small avocado is 106.2 grams.

The minimum weight of a premium avocado is 182.6 grams.

(b) Find the value of μ and of σ . [5]

A supermarket purchases all the avocados from the farm that weigh more than 106.2 grams.

(c) Find the probability that an avocado chosen at random from this purchase is categorized as

(i) medium;

(ii) large;

(iii) premium. [4]

The selling prices of the different categories of avocado at this supermarket are shown in the following table:

Category	Medium	Large	Premium
Selling price (\$) per avocado	1.10	1.29	1.96

The supermarket pays the farm \$200 for the avocados and assumes it will then sell them in exactly the same proportion as purchased from the farm.

(d) According to this model, find the minimum number of avocados that must be sold so that the net profit for the supermarket is at least \$438. [4]

References:

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12EP10

3.

Note: In this question, do not penalise incorrect use of strict inequality signs.

Let X = mass of a bag of sugar

(a) evidence of identifying the correct area **(M1)**

$$P(X < 995) = 0.0765637...$$

$$= 0.0766$$

A1

[2 marks]

(b) 0.0766×100
 ≈ 8

A1

[1 mark]

Note: Accept 7.66.

(c) recognition that $P(X > 1005 | X \geq 995)$ is required **(M1)**

$$\frac{P(X \geq 995 \cap X > 1005)}{P(X \geq 995)}$$

$$\frac{P(X > 1005)}{P(X \geq 995)}$$

(A1)

$$\frac{0.0765637...}{1 - 0.0765637...} \left(= \frac{0.0765637...}{0.923436...} \right)$$

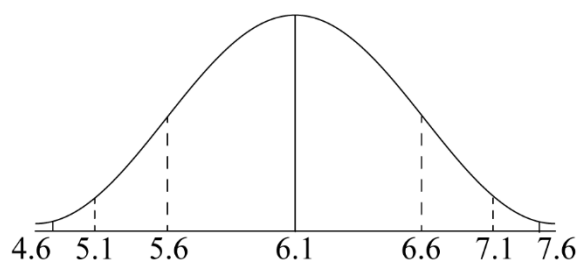
$$= 0.0829$$

A1

[3 marks]

Total [6 marks]

4. (a)



A1A1

Note: Award **A1** for a normal curve with mean labelled 6.1 or μ , **A1** for indication of SD (0.5): marks on horizontal axis at 5.6 and/or 6.6 **OR** $\mu - 0.5$ and/or $\mu + 0.5$ on the correct side and approximately correct position.

[2 marks]

(b) $X \sim N(6.1, 0.5^2)$

$P(5.5 < X < 6.5)$ **OR** labelled sketch of region

$= 0.673$ (0.673074...)

(M1)

A1

[2 marks]

(c) $(P(X < 5.3) =) 0.0547992...$

$0.0547992... \times 80$

$= 4.38$ (4.38393...)

(A1)

(M1)

A1

[3 marks]

(d) 0.15 **OR** 0.85

(A1)

$P(X > x) = 0.15$ **OR** $P(X < x) = 0.85$ **OR** labelled sketch of region

(M1)

6.62 (6.61821...)

A1

[3 marks]

(e) $(P(X > 6.25) =) 0.382088...$

recognition of binomial

e.g. $B(10, 0.382088...)$

0.0502 (0.0501768...)

(A1)

(M1)

A2

[4 marks]

Total [14 marks]

8. (a) use of inverse normal to find z -score (M1)

$$z = 2.0537...$$

$$2.0537... = \frac{82 - 75}{\sigma} \quad (A1)$$

$$\sigma = 3.408401...$$

$$\sigma = 3.41$$

A1

[3 marks]

- (b) evidence of identifying the correct area under the normal curve (M1)

$$P(T > 80) = 0.071193...$$

$$P(T > 80) = 0.0712$$

A1

[2 marks]

- (c) recognition that $P(80 < T < 82)$ is required (M1)

$$P(T < 82 | T > 80) = \frac{P(80 < T < 82)}{P(T > 80)} = \left(\frac{0.051193...}{0.071193...} \right) \quad (M1)(A1)$$

$$= 0.719075...$$

$$= 0.719$$

A1

[4 marks]

continued...

Question 8 continued

(d) recognition of binomial probability **(M1)**

$$X \sim B(64, 0.071193...) \text{ or } E(X) = 64 \times 0.071193... \quad \textbf{(A1)}$$

$$E(X) = 4.556353...$$

$$E(X) = 4.56 \text{ (flights)}$$

A1

[3 marks]

(e) $P(X > 6) = P(X \geq 7) = 1 - P(X \leq 6)$ **(M1)**

$$= 1 - 0.83088...$$

(A1)

$$= 0.1691196...$$

$$= 0.169$$

A1

[3 marks]

Total [15 marks]

9. (a) $P\left(\frac{\mu - 1.5\sigma - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{\mu + 1.5\sigma - \mu}{\sigma}\right)$ (M1)

$P(-1.5 < Z < 1.5)$ OR $1 - 2 \times P(Z < -1.5)$ (A1)

$P(-1.5 < Z < 1.5) = 0.866385\dots$

$P(\mu - 1.5\sigma < X < \mu + 1.5\sigma) = 0.866$ A1

Note: Do not award any marks for use of their answers from part (b).

[3 marks]

(b) $z_1 = -1.75068\dots$ and $z_2 = 1.30468\dots$ (seen anywhere) (A1)

correct equations (A1)(A1)

$\frac{106.2 - \mu}{\sigma} = -1.75068\dots, \mu + 1.30468\dots\sigma = 182.6$

attempt to solve their equations involving z values (M1)

$\mu = 149.976\dots, \sigma = 25.0051\dots$

$\mu = 150, \sigma = 25.0$ A1

[5 marks]

(c) (i) new sample space is 96% (may be seen in (ii) or (iii)) (M1)

$P(\text{medium}|\text{not small})$ OR $\frac{0.576}{0.96}$

$P(\text{Medium}) = 0.6$ A1

(ii) $P(\text{Large}) = 0.3$ A1

(iii) $P(\text{Premium}) = 0.1$ A1

[4 marks]

continued...

Question 9 continued.

(d) attempt to express revenue from avocados (M1)

$$1.1 \times 0.6 + 1.29 \times 0.3 + 1.96 \times 0.1 \text{ OR } 1.243n$$

correct inequality or equation for net profit in terms of n (A1)

$$1.1 \times 0.6n + 1.29 \times 0.3n + 1.96 \times 0.1n - 200 \geq 438 \text{ OR } 1.243n - 200 = 438$$

attempt to solve the inequality (M1)

sketch OR $n = 513.274\dots$

$$n = 514$$

A1

Note: Only award follow through in part (d) for 3 probabilities which add up to 1. FT of probabilities from c) that do not add up to 1 should only be awarded **M** marks, where appropriate, in d).

[4 marks]

Total [16 marks]